The Cost of Ignoring Compositional Geometry Across Scales

Quantified Errors from EITT as a Diagnostic Framework

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The Central Claim

Every time a scientist averages, aggregates, or compares compositional data across different scales without respecting the Aitchison geometry of the simplex, they introduce systematic errors. These errors are not noise. They are structural, predictable, and quantifiable.

EITT (Entropy Invariance under Temporal Transformation) was designed to test whether information is preserved when compositional time series are compressed across temporal scales. What it became was a general-purpose error detector for scale-crossing operations on the simplex. Every failure mode diagnoses a specific violation of compositional geometry. Every success confirms that the geometry was respected.

HUF is inert. It does not transform data. It holds a mirror up to a system and measures the gap between what the system thinks it is and what the geometry says it is. The errors were always there. HUF just made them readable.

1. Temporal Scale: Averaging Compositions Over Time

The most common operation in applied science is temporal averaging: monthly to yearly, daily to weekly. When the data are compositions, the arithmetic mean is the wrong operation. It does not live on the simplex. The correct operation is the geometric mean (Frechet mean under the Aitchison metric), which corresponds to the arithmetic mean in CLR coordinates.

Geometric-mean decimation (CoDa-correct):

Dataset	D	N	delta 2:1	delta 5:1	delta 10:1
Gold/Silver prices	2	339	+0.08%	+0.53%	+1.39%
World electricity	7	25	-0.59%	-0.09%	-3.15%
Financial sectors	9	74	+0.004%	-0.008%	+0.01%
Germany electricity	7	36	+0.03%	-1.26%	-4.37%

At 2:1 compression, errors are under 0.6% in every domain. Information is preserved when averaging respects the geometry.

The gap when you use arithmetic means instead (CoDa-incorrect):

Dataset	Scale	Geom delta	Arith delta	Gap
Germany electricity	5:1	-1.26%	+0.27%	**1.53 pp**
Germany electricity	10:1	-4.37%	-3.24%	**1.13 pp**
Gold/Silver	10:1	+1.39%	+1.74%	0.35 pp

Every climate model that averages monthly energy generation shares to yearly using arithmetic means carries this error. It is small at short scales and dangerous at long ones - exactly the regime where climate and economic analyses operate.

2. Dimensional Scale: Amalgamating Parts Hides Structure

Scientists routinely combine compositional parts for simplicity. Seven energy sources become three: 'Fossil, Nuclear, Renewable.' This is amalgamation. When done carelessly, it hides internal structure that affects the behaviour of the composition. EITT Inversion detects this.

China electricity (Ember data, 2000-2024) - DOWNWARD inversion:

Composition	K	delta 2:1	delta 3:1	Status
Full (all 9 fuels)	9	**-2.22%**	**-2.33%**	**FAIL**
Fossil / Nuclear / Renewable	3	-1.13%	-1.12%	PASS
Fossil / Non-Fossil	2	-0.90%	-0.89%	PASS

Gold/Silver (1688-2026) - UPWARD inversion:

Composition	K	delta 365:1	Status
Gold share / Silver share	2	**+6.7%**	**FAIL**
+ Volatility + Momentum carriers	4	+0.38%	PASS

Inversion is bidirectional. **Upward** (gold/silver): too few dimensions hide structure - increase K to restore invariance. **Downward** (China): too many dimensions expose non-stationarity - decrease K to mask it. The direction of the fix is itself diagnostic.

In 2000, amalgamating wind and solar into 'Renewable' was harmless (0.2% of mix). In 2024, they are 14.7% and growing exponentially. The same amalgamation now hides 2.22% of entropy error. The dimensional scale of the composition must evolve with the system it describes.

3. Metric Scale: Distances Expand, Entropy Holds

Common assumption: temporal averaging smooths things out. Points get closer together. On the simplex, this assumption is wrong.

Aitchison distance expansion ratios (consecutive observations):

Dataset	2:1	3:1	5:1
Gold/Silver	1.32x	1.51x	1.95x
Energy World	**2.01x**	**3.04x**	**5.08x**
Financial	1.44x	1.64x	2.00x
Germany	1.40x	1.81x	2.17x

Distances grow 2-5x under averaging. Decimation is NOT a contraction mapping on the simplex. Yet entropy is preserved (delta < 0.6% at 2:1). The operation preserves information content while expanding the metric structure. Anyone assuming contraction is using Euclidean intuition that does not transfer to the simplex.

4. Functional Scale: It Is Not About Shannon

We tested the entire Renyi family H_q from q = 0.1 to q = 5.0, and the Tsallis family at the same values:

Renyi entropy residuals delta at M=2:1:

q	Energy World	Gold/Silver	India
0.1	-0.15%	-	-
0.5	-0.33%	+0.13%	-0.12%
1.0 (Shannon)	-0.42%	+0.19%	-0.02%
2.0 (Herfindahl)	-0.45%	+0.20%	+0.21%
5.0	-0.49%	-	-

Every single q-value passes. Shannon is not special. The invariance lives in the Aitchison geometry, not the entropy functional. This means every smooth functional on the simplex - including Simpson's diversity (ecology), the Herfindahl index (economics, q=2), Gibbs entropy (physics) - will exhibit the same near-invariance when CoDa-correct averaging is used, and the same errors when it is not.

5. Stationarity Scale: The Boundary Where Everything Breaks

The deepest error is temporal: assuming that a process stationary at one scale remains stationary at another. d(CoDa)/dt - the maximum balance velocity across ILR coordinates - is the control signal that detects this.

Failure	What Broke	delta	Diagnosis
World energy 10:1	Block spans structural change	-3.15%	10 years of transition averaged as stationary
Germany 10:1	Nuclear phase-out	-4.37%	35 years of Energiewende in one average
Renewables-only 2:1	Exponential solar/wind	-4.51%	Non-stationary subcomposition
China K=9 at 2:1	Individual fuel volatility	-2.22%	Too many non-stationary dimensions
Solar|Rest balance	Solar growth rate	shift -0.12	Fastest-moving part breaks one tap

Adaptive response: d(CoDa)/dt as the tap controller:

Dataset	Fixed delta	Adaptive delta	Compression
Gold/Silver (10:1)	+1.39% (marginal)	-0.43% (pass)	10.0:1
Solar|Rest	shift -0.12 (fail)	shift +0.08 (pass)	3.6:1
Germany	all fail	all fail	no safe window

When composition changes slowly, compress aggressively. When fast, use smaller blocks. When fast everywhere (Germany), no averaging is safe. The failure itself is the diagnostic.

6. The Complete Error Catalogue

Scale Violation	Error Source	Measured Magnitude	Who Is Affected
Arithmetic mean across time	Wrong averaging operator	**Up to 1.53 pp** excess error	Climate, economics, ecology
Amalgamating volatile parts	Hidden non-stationarity	**2.22%** entropy error	Energy policy, portfolios
Insufficient dimensions	Missing compositional structure	**6.7%** entropy error	Any analysis choosing K by convention
Assuming contraction	Euclidean intuition on simplex	Distances grow **2-5x**	Signal processing, time series
Fixed window on non-stationary data	Ignoring d(CoDa)/dt	**4.51%** error	Any long-horizon forecast
One entropy functional	Missing generality	ALL q in [0.1, 5.0] pass	Information theory, ecology, econ

Every number comes from a reproducible test. Code, data, and results in the repository.

7. What CoDa Provides That Nothing Else Does

The **geometric mean** is not just 'another kind of average.' It is the Frechet mean under the only metric that respects the constraint structure of compositions. Using it eliminates the temporal averaging error.

The **ILR transform** is not just 'another coordinate system.' It is the isometric bridge between the simplex and unconstrained Euclidean space. Derivatives, variances, and distances computed in ILR coordinates are geometrically correct. Computing them on raw shares is not.

Subcompositional coherence guarantees that your analysis does not change its answer when you look at a subset of parts. L2 does not have this property. Aitchison distance does.

The **SBP-invariance of total variance** means that no matter how you wire the balance transformer, the total information content is the same. Our multi-tap test (213 measurements, 8 SBP trees, 94.8% pass) confirms this. The 5.2% that fail are diagnostic: they identify Solar (too fast) and Nuclear (structural break).

The Aitchison geometry is not a correction you apply to data. It is the geometry the data already lives in. When you ignore it, you get errors. When you respect it, those errors disappear. The errors were never in HUF. They were in the assumption that the simplex is flat.

All code, data, and test results: github.com/PeterHiggins19/Higgins-Unity-Framework
Multi-AI adversarial review (Claude, ChatGPT, Grok, Gemini, Copilot) applied throughout.
Tested on: 4 domains | 3 countries | 339 years commodity + 25 years energy + 74 months financial | Renyi q = 0.1 to 5.0 | 8 SBP trees | 213 balance measurements